Renewable Energy 179 (2021) 705–722

Contents lists available at ScienceDirect

Renewable Energy

journal homepage: www.elsevier.com/locate/renene

A biogas-solar based hybrid off-grid power plant with multiple storages for United States commercial buildings

Barbara Mendecka *, Daniele Chiappini , Laura Tribioli , Raffaello Cozzolino

Department of Industrial Engineering, University of Rome Niccolò Cusano, Italy

A R T I C L E I N F O

Article history:
Received 5 March 2021
Received in revised form
8 July 2021
Accepted 18 July 2021
Available online 20 July 2021

Keywords:
Biogas
Hybrid renewable energy
Microgrid
Polygeneration plant
Solar energy
Unitized regenerative fuel cell

A B S T R A C T

In this paper, a hybrid renewable power plant with a storage system is designed. The benefits of sizing and energy management are assessed for a commercial building under eight different climatic conditions in the United States. In the considered system, photovoltaic panels are coupled to a unitized regenerative solid oxide fuel cell. The use of biogas to feed unitized regenerative solid oxide fuel cell is investigated, employing a detailed electrochemical model of electrolyzer and fuel cell modes. A battery pack is included in the plant as a secondary storage system, together with a diesel engine operating in backup mode.

Four scenarios where biogas amount is varied together with the initial state of charge of the battery were evaluated. Results demonstrate that the power plant can operate with 100 % renewable procurement if the digester produces from 6000 to 9500 stdm^3/y and the battery is completely charged at the beginning of the year. By reducing the biogas availability or starting with a low state of charge, the use of the diesel generator is inevitable.

The study confirms that the proposed hybrid renewable power plant is technically feasible and can be considered a reliable and clean energy source in other areas and buildings.

1. Introduction

Commercial structures represent approximately 30% percent of the building stock in the United States (U.S.) [1]. Commercial end-use energy consumption depends on size, principle activity, and geographical zone. According to the 2019 annual energy consumption data of the U.S. Energy Information Administration (EIA), it accounts for around 19 PJ (18 trillion Btu) that represents 18% of primary energy consumption in the U.S [2].

As reported by the Commercial Building Energy Consumption Survey (CBECS2018) [3], the commercial buildings sector in the United States is getting larger, and the continuous increase is driven by floor area and population expansions. The total amount of floor space of 5.9 million commercial buildings, larger than 1000 square feet, increased by 11 % from 2012 to 2018 in the U.S. Among the principal building activities, mercantile activity with shopping malls comprised of multiple connected establishments (strip malls) accounts for 9% of all commercial buildings and 11 % of total commercial floor space. More than 70 % of the building stock was constructed before 2000 when energy use and environmental impact due to the design and construction of buildings was still considered a minor issue [4,5].

End-use energy consumption and on-site generation in the commercial sector are also expected to grow continuously. Even if some energy reduction strategies have already been implemented, leading to emissions leveling off in the period from 2013 to 2016, the trend of carbon dioxide pollution started increasing again up to 9.7 Gt and around 40 % of total carbon dioxide emissions in 2018 [1,6]. The electricity consumption in buildings is strategically essential to target for energy and emission reduction, as it contributes to over 70 % of primary energy consumption [7].

Implementing measures to improve energy efficiency and mitigate building stock emissions is no longer a choice but rather a necessity. An effective measure that allows achieving both economic, energy, and air quality goals is to increase the share of renewable energy sources (RES) in the building sector. Specifically, a shift to RES-based microgrids with distributed energy resources (DERs) represents an effective way to reduce the environmental externalities and decarbonize the building sector, and at the same time to increase the grid resiliency [8]. DERs microgrid forms a

* Corresponding author.
E-mail address: barbara.mendecka@unicusano.it (B. Mendecka).

https://doi.org/10.1016/j.renene.2021.07.078
0960-1481/© 2021 Published by Elsevier Ltd.

Nomenclature

AC	alternating current
BAT	battery
BGG	biogas combustion engine
BMG	biomass gasifier
BoP	balance of plant
CBECS	Commercial Building Energy Consumption Survey
COMP	compressor
CONV	converter
DER	distributed energy resource
DC	direct current
DG	diesel generator backup
EIA	Energy Information Administration
EL	electric
FC	fuel cell
HRE	hybrid renewable energy
ICE	internal combustion engine
L	load
LCOE	levelized cost of electricity
NREL	National Renewable Energy Laboratory
PEM-URFC	polymer electrolyte membrane fuel cell
PHS	pumped hydro storage
PV	photovoltaic
RES	renewable energy sources
RFC	reversible/regenerative fuel cell
SOC	state of charge
SOFC	solid oxide fuel cell
URSOFC	unitized regenerative solid oxide fuel cell
X	specific parameter
VRES	variable renewable energy sources
WT	wind turbine

Subscripts and superscripts

i	flow
k	component

single controllable structure interconnected with an electric load that can operate in grid-connected (on-grid) or isolated (off-grid) mode [9].

Among various RES, solar photovoltaic (PV) technology is the primary driver of RES-based microgrid and off-grid generation. It is considered one of the fastest-growing, most mature, and cost-competitive renewable energy technologies [10]. Moreover, implementing on-site generation with solar PV is essential for reaching Zero-Energy Building (ZEB) goal [11].

Standalone PV is variable, non-controllable, and limited DER. Thus it may not ensure a provision of continuous load and reliable performance of the isolated energy system. Variable RES (VRES), notably solar photovoltaics, can be converted into the dispatchable generation when integrated into an energy storage system (EES). Such a configuration allows to maintain and control the balance between generation and load and increase the overall RES deployment in the system and reduce operational energy costs [12]. However, utilization of single renewable generation technology, even dispatchable, usually requires a backup generator. The general trend is to use a conventional diesel generator combined with variable RES to meet a peak load [13].

Nowadays, hybrid renewable energy (HRE), which is a combination of dispatchable RES including variable and inherently dispatchable RES (IRES) such as bioenergy and geothermal, is becoming a preferable alternative to single RES based on a standalone energy system with conventional backup and storage [14,15]. Moreover, recent studies demonstrate that the costs of HRE systems will continue to decline, making renewables more competitive at the utility scale [16]. Besides, the development of the HRE system plays a significant role in the transition to a 100 % renewable operation [17,18].

Previous works in the literature address the issues such as techno-economic performance, adequate utilization of the resources, and design and size optimization of the HRE systems in different geographical contexts.

For this study, a comprehensive survey on recent case studies of HREs focused on off-grid systems has been conducted and summarized in Table 1. The table lists the modeled HRE technologies, load type, and geographical context. Besides, the main findings of each study are highlighted in the last column of the table.

From the analyzed literature, it can be noted that there are many configurations of HREs that can be successfully implemented to meet the local demand. The combination of solar and wind with conventional diesel generator backup is the most popular configuration studied, as also confirmed in Ref. [40]. However, it has been recently demonstrated that integrating solar and biomass as baseload sources is the most effective configuration, especially for remote areas and decentralized applications.

From the available research, it is also evident that there is no single optimal hybrid energy system configuration. However, system sizing, optimal configuration, and control approach should be appropriately addressed during the design and implementation of HRE. The optimal configuration depends on the studied area (weather conditions, availability of renewable and non-renewable resources), electric load patterns as well as financial and environmental policies.

It is also noted that major studies have evaluated the techno-economic feasibility of HREs systems for electrification of rural areas with specific electric load patterns in a single geographical context. Moreover, a reliable HRE system should also employ efficient control strategy solutions to manage system stability and flexibility. Therefore, implementation of an optimal energy management strategy is essential to optimize the techno-economic performance of HREs systems.

In addition to this, one can observe that high-efficiency energy storage systems, including batteries and/or hydrogen storage systems, must be incorporated in the system to tackle the problem of renewable sources instability and ensure balancing electrical demand and surplus energy production. Finally, from the analysis, it emerges that a limited number of studies have covered HREs that includes fuel cell as a storage and backup system, as evident from Table 1.

Hydrogen as an energy carrier and fuel cells (FC) technologies are becoming increasingly valuable as their needs and benefits within the distributed generation sector are identified [41,42]. Hydrogen can be stored for extended periods and in large capacities. Their main limitations, though, are high economic cost, low round trip efficiency of H_2, and production and storage safety issues connected to the risk of leakage [43]. For hydrogen storage and utilization in the building sector, separated electrolyzer and fuel cell are usually employed [44]. Facci et al. [45] studied the techno-economic feasibility of a solid oxide fuel cell (SOFC) based system for cooling, heating, and electricity production for residential applications. They found all the considered SOFC configurations significantly reduced the yearly energy cost with respect to the usual scenario, where electricity is only acquired from the grid,

Table 1
Summary of grid-isolated HREs (PV-photovoltaics, WT-wind turbine, BGG-biogas combustion engine, BMG-biomass gasifier, BAT-battery storage, H_2-hydrogen storage, PHS-pumped hydro storage, DG-diesel generator backup, FC-fuel cell backup).

Ref	Year	Location	Load profile	HRE components				Main findings
				VRES	IRES	EES	Backup	
[19]	2021	Auniati Satra, India	Rural off-grid area: Diary farm	PV	BGG	BAT		Higher LCOE for the HRE than for PV standalone system. Better CO_2 mitigation for HRE with respect to a standalone system.
[20]	2021	Tamil Nadu, India	Educational building	PV	–	BAT	DG	Development of an approach for optimal sizing of HRE for the specific case study in single building application
[21]	2021	Kai Gan Qi Country, China	Rural off-grid area: domestic applications	PV	BGG	BAT	DG	Development of an approach for optimal sizing of HRE for the specific case study in the rural application
[22]	2021	n/a	Residential building	PV|WT		BAT	DG	Confirmation that electricity excess is the cheapest source for the generation of heat in the system.
[23]	2021	Morocco	Rural off-grid area: residential community	PV	BGG	BAT		Demonstration of economic feasibility of hybrid PV/biogas power generation in designing 100 % electric rural houses
[24]	2021	Cyprus	Educational building	PV|WT		PHS|BAT		Feasibility of the proposed HRE system and possibility to meet close to 100 % of the energy demand.
[25]	2020	Cuenca-Ecuador	Educational building	PV	BMG	BAT		Demonstration of techno-economic feasibility of the off-grid HRE.
[26]	2020	Karnataka, India	Rural off-grid area Cluster of villages	PV|WT	BMG|BGG	H_2|BAT	FC	Demonstration that PV|WT|BMG|BGG|FC|Bat is best-suited, among the analyzed, HRE configuration.
[27]	2020	Australia	Off-grid area: residential community	PV		H_2|BAT	DG	Demonstration that the H_2|Bat hybrid energy storage system is the most cost-effective scenario, though all developed scenarios.
[28]	2019	Dongola, Sudan	Off-grid area: residential community	PV|WT		BAT	DG	Confirmation of cost-effectiveness and reliability of proposed HRE in the specific scenario.
[29]	2019	Shibpur, India	Educational building	PV|WT	BGG	BAT		Experimentally validated model of HRE. Demonstration of project viability and the possibility of upscaling.
[30]	2018	Sabah, Malysia	Rural off-grid area	PV		BAT	DG	Demonstration of the high flexibility of selected HRE system overall conditions
[31]	2018	Turkey	Off-grid area: residential community	PV|WT		H_2|BAT		The lowest LCOE result for PV/WT/H_2/DG system. H_2 storage is more costly than battery storage.
[32]	2017	Katakhali, Bangladesh	Rural off-grid area	PV|WT	BGG	BAT	DG	Confirmation of economic feasibility of proposed HRE in optimized configuration compared to diesel-only systems. The proposed system has reasonable environmental benefits and an attractive payback period.
[33]	2017	Tezpur, India	Rural off-grid area	PV	BMG	BAT	DG	Feasibility and reliability of HRE in energy supply to the remote village.
[34]	2017	South China Sea, Malaysia	Resort center	PV|WT		BAT	DG	Confirmation that the optimized system has better economic and environmental performance than the diesel-only system.
[35]	2017	Kallar Kahar	Rural off-grid area	PV|WT	BMG			Demonstration that the cost of the optimal configuration of the HRE system falls into an acceptable range for a renewable hybrid system.
[36,37]	2016	Mediterranean site	Artificial islands	PV|WT		H_2|BAT	DG|FC	Demonstration of an economically efficient mechanism to generate RES-based electrical power in isolated systems
[38]	2016	Bardsir, Iran	Rural off-grid area	PV	BGG	H_2	FC	Demonstration of optimized, and validated model of an HRE system FCs can be promising energy generation systems for electrification to standalone applications in comparison with the other energy sources.
[39]	2016	Mazara del Vallo, Italy		PV|WT		H_2|BAT	DG|FC	Demonstration of SOFC-fed by biogas, integrated into a hybrid renewable power plant, for an off-grid application demonstrating that power plant can achieve 100 % renewable operation.

thermal energy is produced through the boiler, and cooling energy is obtained from a mechanical refrigerator.

A unitized regenerative fuel cell (URFC) employment rather than separate units may represent an advancement, especially in a space-constrained application. It has been demonstrated that the employment of unitized regenerative polymer electrolyte membrane fuel cell (PEM-URFC) in HRE is beneficial from economic and environmental points of view, as reported in Refs. [15,46]. Photovoltaic panels may also be coupled to a unitized regenerative solid oxide fuel cell (URSOFC) as a storage/backup system and with an anaerobic digester for biogas on-site production and utilization in SOFC as proposed in Ref. [47].

Despite many investigations on photovoltaic (PV)/wind/fuel cell (FC) systems, the integration of PV/URSOFC fed by biogas is rarely found. Moreover, most of the studies focus on off-grid rural application in one specific geographical context.

This study estimates the energetic effectiveness of an off-grid hybrid RES power plant to meet the electrical load of a commercial building across different climate zones in the U.S.

Specifically, deployment of local biogas, hydrogen production, and the integration of PV/URSOFC systems, aimed at 100 % renewable operation mode integration, is investigated, including a sensitivity analysis on the biogas production rate and battery state of charge. In addition, different geographical contexts are studied to explore the differences between both supply and load patterns.

In order to contribute to the existing body of literature in the HRE domain, the following specific issues are addressed: i) development of a hybrid solar-based power plant concept to maximize RES penetration, including a comparison of the overall efficiency of the system working in different U.S. locations, ii) in-depth description of the optimization procedure followed for the plant sizing iii) application of validated and optimized system operation strategy in order to manage cooperation between various RES, iv) reversible operation of URSOFC with both hydrogen and biogas v) investigation of the effects of the different initial state of charge levels in the battery and different biogas production rates, vi) demonstration of the possibility of 100 % RES operation.

The remaining part of the paper is organized as follows: case study definition and overall methodological framework used in this paper are reported in Section 2. The results of the study are discussed in Section 3. The concluding remarks are presented in Section 4.

2. Materials and methods

2.1. Hybrid renewable energy system

The proposed hybrid RES-based power plant, shown in Fig. 1 consists of a photovoltaic panel array (PV), a battery pack (BAT), a unitized regenerative solid oxide fuel cell (URSOFC) wired into DC subarray, AC/DC inverter, and a diesel backup generator (ICE).

The available is always used as a primary power supplier. URSOFC and the battery pack are aimed at satisfying the load - when the power provided by the solar power generator is not sufficient. When excess power from renewable energy sources is available, both: battery and hydrogen tank are charged. In addition to that, to further reduce the plant environmental impact, an anaerobic digester is considered, which can be fed by food waste locally produced. Such a solution allows mitigating the negative environmental impact of waste transportation to large processing facilities, and at the same time, feeding the URSOFC with the produced biogas. For this reason, the choice of a SOFC is made due to the possibility of directly operating with biogas from anaerobic digestion [48,49]. The rate of biogas production varies over the day, and the use of the fuel cell in reverse mode can further increase the fuel availability without the need of an additional component, i.e., electrolyzer. The unitized regenerative solid oxide fuel cell model is integrated with all the needed auxiliary components for the necessary balance of power, and the results have been implemented in the power plant model to properly simulate power/hydrogen production and consumption at different loads. Conversely, the ICE is used only to meet the primary load in the event of a lack of energy from the other sources. Further, electricity generated from PV and fuel cell is converted to AC to supply commercial building while DC power is used for supplying electrolyzer to generate hydrogen.

Referring to Fig. 1, the energy balance of the system can be expressed as follows [47]:

$$P_L = P_{PV} + P_{FC} - P_{EL} + P_{BAT} + P_{ICE} \tag{1}$$

where P_L is the net load power demand, given by the end-user power demand minus the power required by the biogas and hydrogen compressors (P_{COMP}) and power losses in rectifiers and converters (P_{CONV}), P_{PV} is the renewable power supplied by the PV array, P_{FC} and P_{EL} are the power supplied by the URSOFC in FC-mode and the power absorbed by the URSOFC in EL-mode, respectively, P_{BAT} is the battery power, positive in discharge and negative in charge, P_{ICE} is the power provided by the engine.

The operational stage of the system has been modeled by applying the power management strategy described in the next paragraph. Battery state of charge (SoC) dynamics and the variations of level and composition of the biogas/hydrogen mixture have been modeled using the MATLAB/Simulink® simulator employing an energy-based quasi-static approach [14]. PV system and ICE have been modeled employing phenomenological, semi-empirical models or stationary maps as described in detail in previous works [15]. Biogas is considered as an input flow that can be produced locally in the anaerobic digester.

2.2. Unitized regenerative solid oxide fuel cell modeling

The URSOFC electrochemical model is based on a model already validated in Ref. [49] and is briefly presented here. The URSOFC is ruled by the following reactions (2) occurring at the cathode and anode side, respectively:

$$\begin{gathered} O_2 + 4e^- \rightarrow 2O^{2-} \\ 2H_2 + 2O^{2-} \rightarrow 2H_2O + 4e^- \end{gathered} \tag{2}$$

Considering ohmic, activation, and diffusion losses, ξ_{ohm}, ξ_{act} and ξ_{dif} the load voltage V_{URSOFC} can be calculated from the Nernst potential, V_{Nerst}, based on the following Eq. (3):

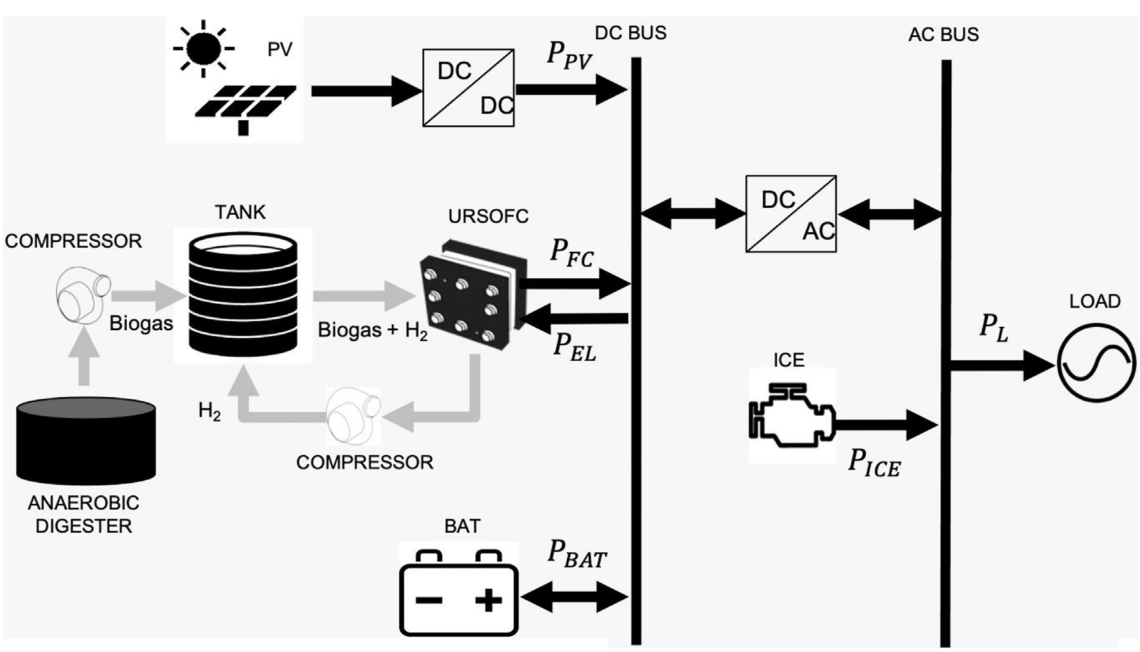


Fig. 1. Schematic configuration of an off-grid hybrid renewable energy system.

$$V_{URSOFC} = V_{Nerst} + k\left(\xi_{ohm} + \xi_{act} + \xi_{dif}\right) \quad (3)$$

with:

$$k = \begin{cases} 1 & \text{EL Mode} \\ -1 & \text{FC Mode.} \end{cases}$$

Where EL is the electrolyzer mode and FC is the fuel cell mode. These losses are a function of the circulating current, which affects the inlet/outlet compositions, and an in-depth explanation of all these terms is available in Ref. [48] and is here avoided for the sake of fluency.

The current density i is defined according to the number of hydrogen molecules flowing at anode side and effectively converted, as per Eq. (4):

$$n_{H_2} = \frac{i}{2F \cdot U_F} \quad (4)$$

In this paper, the choice is made to use a unique tank for both the biogas and the hydrogen produced by the electrolyzer. This is due mainly to space requirements rather than investment costs. Therefore, the URSOFC is fed by a mixture of biogas (60 % CH_4 and 40 % CO_2 in vol., as in Ref. [39]) and pure hydrogen (100 % H_2) at different ratios, varying according to the power plant operating conditions. The URSOFC performance, in terms of polarization curves for a single cell, have been derived by changing the share of inlet pure hydrogen and biogas masses. The obtained curves presenting cell voltage behavior against current density for a single cell, used in the power plant model, are reported in Fig. 2:

2.3. The control strategy

The plant power management has been achieved through a rule-based approach, which guarantees reliability and computational effectiveness despite its simplicity. This approach has been previously developed, optimized, and validated in Refs. [14,15,50], and the flow chart summarizing its rules is given in Fig. 3.

As one can note from Fig. 3, the primary power supply source is represented by the solar energy (P_{PV}), whereas the URSOFC (P_{FC}), together with the batteries (P_{BAT}), represent secondary sources that can be used to fulfill instantaneous power demand (P_L) – in the case where the solar energy cannot reach the requested target – or, in addition, they can be used to store the excess power when available.

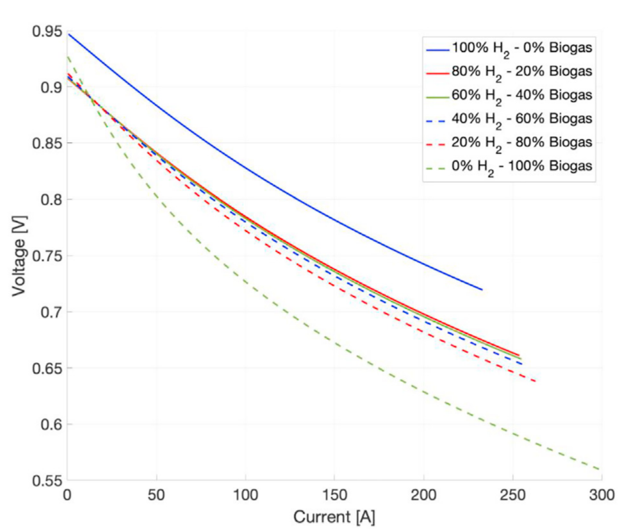


Fig. 2. Current density [A] vs cell voltage [V] for different inlet mixtures on URSOFC performance.

In “Power Supply” mode (right dot-dashed box in Fig. 3), when the primary load request overtakes the power supplied by the PV panels, the URSOFC acts like a FC (P_{FC}) until the tank level is greater than its minimum imposed threshold (TL > TL_{MIN}). In the occurrence of low tank value, TL ≤ TL_{MIN}, or in the case of power demand to URSOFC greater than its maximum value in FC-mode ($P_{FC,MAX}$), the battery provides the extra power if the SoC is higher than the minimum threshold (SoC > SoC_{MIN}), so to meet the load target. Finally, the presence of a backup diesel engine is considered in the power plant as well (P_{ICE}), in order to face up with power demands to the battery higher than the maximum power suppliable in discharging mode ($P_{BAT,MAX,D}$) from the battery and/or from the URFC ($P_{FC,MAX}$), or in the unwanted condition where both the tank and the battery are empty (TL ≤ TL_{MIN} and SoC ≤ SoC_{MIN}).

Conversely, a relay can rule two different “Power Recovery” modes (right dashed boxes in Fig. 3):

1) The “Power Recovery” mode a) is applied whenever the SoC of the battery falls in the considered SoC range - from SoC = SoC_{MAX} (switch on point) until SoC ≤ SoC_{SUST} (switch off point, equal to 50 %) -. The URSOFC acts in the EL-mode (P_{EL}) as a primary energy recovery apparatus, with the aim of producing and storing hydrogen to refill the tank up to its maximum threshold (TL < TL_{MAX}). In addition, the battery, until charging is allowed (SoC < SoC_{max}), can absorb the extra power coming from the PV panels when the maximum URSOFC-EL-mode power is reached ($P_{EL,MAX}$). Finally, the extra power (P_{EXCESS}) is dumped when the power demand to the battery is higher than the maximum power absorbable in charging operation ($P_{BAT,MAX,C}$) or both battery and tank are full (TL > TL_{MAX} and SoC > SoC_{MAX}).
2) The “Power Recovery” mode b) starts when the state of charge drops below the sustenance value SoC = SoC_{SUST} (switch on point). The battery represents the primary energy recovery system in this mode until it is fully charged (SoC = SoC_{max}, switch off point). Again, if the tank is not completely filled (TL < TL_{MAX}), the URSOFC can absorb the surplus arising from the unbalance in between the power provided by the PV panels and the one which can be absorbed from the battery in charging mode ($P_{BAT,MAX,C}$). Finally, another dump load is considered for this mode as well, whenever both tank and battery are full (TL > TL_{MAX} and SoC > SoC_{MAX}) or the power demand to the URSOFC is higher than the maximum power it can absorb ($P_{EL,MAX}$).

The switching between different “Power Recovery” modes was aimed at balancing the amount of energy recovered between the fuel tank and the battery to reduce the extra energy dumped at the end of one year of operation. In the pursuit of that, the value of SoC_{SUST} has been set equal to 50 %, as a result of a sensitivity analysis carried out under this constraint after the plant sizing was defined.

As stated above, diesel engine usage should be avoided for environmental aspects. Thus, it is used to meet the primary load request in the unwanted occurrence of lack of energy from all the other sources, having imposed that both hydrogen tank and the battery should be charged only from renewable sources.

2.4. Case study and hybrid power plant sizing

The U.S. Department of Energy (DOE) Building America Program developed a report with 16 commercial reference buildings covering approximately 70% of the total commercial buildings in

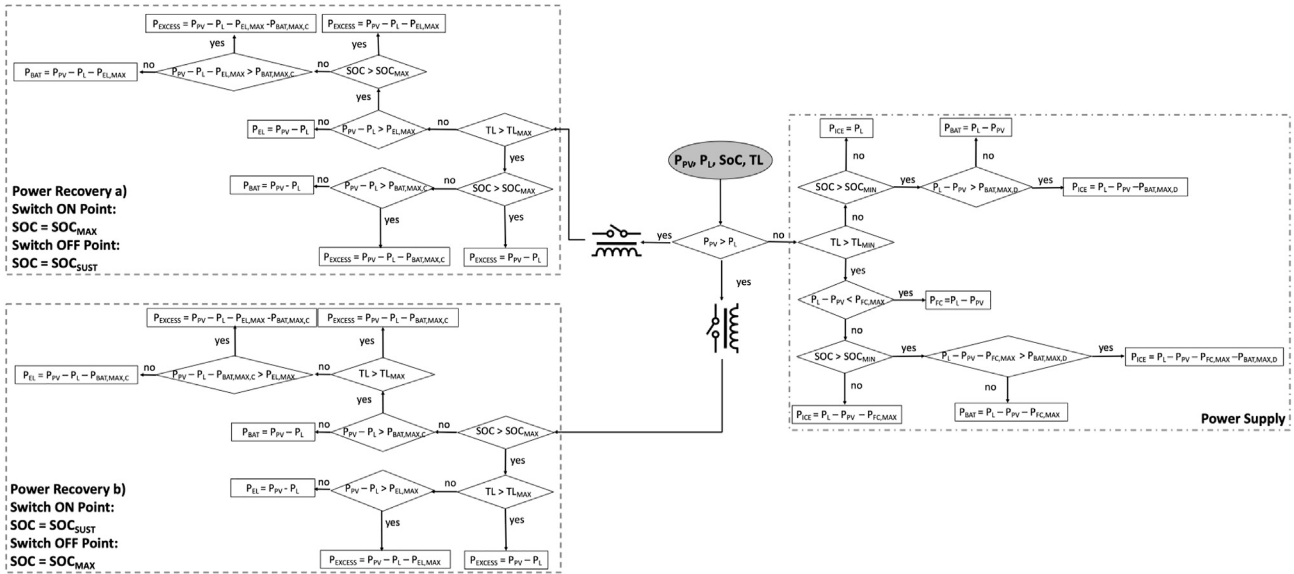


Fig. 3. Control strategy flow chart.

different locations of all United States climate zones [51]. The report contains a complete description of the structural and shape characteristics as well as data related to the energy needs by end-uses, energy sources, and CO_2 emissions of each commercial reference building.

Among the 16 reference commercial buildings, in this study, the strip mall was examined for its structural configuration, particularly favorable for the installation of photovoltaic panels and for the high electrical needs (cooling, interior, and exterior lighting, interior equipment, fans) to be satisfied.

The U.S Building America Program climate zones, defined in the report [51] and considered in this study, are derived from climate designations used by the International Energy Conservation Code (IECC) and American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) [52]. According to the IECC map, the United States is divided into eight temperature-oriented climate zones. In turn, these zones are further subdivided into three designated humidity regimes A, B, and C. Thus, the IECC map foresaw up to 24 potential climate designations. The climate zone definitions are based on heating degree days, average temperatures, and precipitation. For simplifying representative purposes, these are combined into the macro zones Hot-humid, Hot-dry/mixed-dry, Mixed-humid, Marine, Cold/very cold, and Subarctic, as presented in Fig. 4.

In this analysis, one or more representative cities were considered in each of the five macro climate zones, omitting only the subarctic climate zone due to the improbability of finding a strip mall in these areas. The cities selected for this analysis are presented in Table 2, highlighting the Building America climate index and the respective IECC.

For each city, the hourly solar radiation distribution and air temperature at each latitude and longitude were calculated based on data retrieved from the NASA database [53].

The monthly average daily radiation and temperature value for each city are shown in Fig. 5. Again, it is possible to note a more uniform distribution of solar radiation throughout the year in the hot-humid climatic zones - Miami and Houston -, while higher absolute values are recorded in the hot-dry climatic zones - Los Angeles and Las Vegas.

As previously mentioned, in this study, the strip mall was considered as the reference building suitable for hosting PV rooftop system, whose structural and shape characteristics are shown in Table 3. The building consists of a single floor, with a flat roof area of 2090 m^2, ideal for roofing with a photovoltaic array and steel-frame external walls.

The hourly electrical load of the strip mall was determined using the HOMER Energy software [54] on the basis of the energy consumption assumptions of the U.S Building America Program report and according to each of the cities considered in the analysis. Electricity load profiles are, therefore, a function of the building's end-use and geographic location. For representative simplicity, Fig. 6 shows the load profile obtained for each of the cities involved in the analysis, reporting the monthly data distribution through the minimum and maximum value, the first and third quartile, and the median of each month. However, hourly-based load variation has been considered in the simulation.

The total annual electricity needs by end-uses - cooling, interior and exterior lighting, interior equipment, fans - were also determined. From Fig. 7, it can be seen that for all the cities, the greater weight is due to interior lighting, from 36 to 50 % of the total energy needs. An interesting factor to consider is the electricity consumption for cooling, which for the hot-humid climate zone - Miami and Houston - reaches 27 and 21 % of the total demand, respectively (see Fig. 7).

Further, the power plant has been sized based on the following main assumptions:

1) The diesel generator is used as a backup system. Hence, it is not included in the optimization, and it is sized so as to satisfy the estimated maximum load of each location in case of unavailability for failure or lack of stored energy in the battery or hydrogen for the URSOFC;
2) For the battery, a Hoppecke OPzS solar power vented lead-acid battery module with 3028 Ah @ 5C, and 1.84 V of nominal voltage has been chosen. Each battery module is composed of 26 cells in series, for a total voltage of 48 V. In order to have a 240 V pack, for all the locations, 5 battery modules have been connected in series;
3) The maximum rate of biogas considered in this study is of 30 stdm^3/day, averaged on a monthly basis. This is considered to be

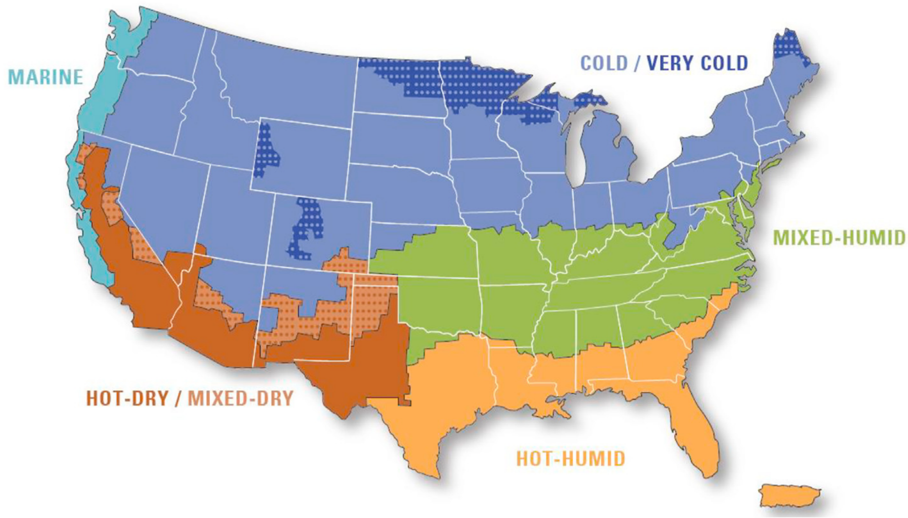


Fig. 4. The U.S. Building America climate zone [52].

Table 2
Cities in different climate zones.

State	City	Building America Climate Zone	IECC Climate Zone	IECC Moisture Regime
Minnesota	Minneapolis	Cold	6	A
Texas	Houston	Hot-Humid	2	A
Nevada	Las Vegas	Hot-Dry	3	B
California	Los Angeles	Hot-Dry	3-coast	B
Florida	Miami	Hot-Humid	1	A
New York	New York	Mixed-Humid	4	4
Colorado	Denver	Cold	5	B
Washington	Seattle	Marine	4	C

given by a food waste production of two medium and three small restaurants, at an average rate of 290 kg/day, with a conversion factor of 0.104 stdm^3 of biogas per kilogram of food waste [55].

For the sizing of the photovoltaic panels and URSOFC, a full factorial design has been performed. All the system simulations have been carried out through an in-house dedicated solver developed in the Matlab/Simulink® environment. The components have been implemented with referring to the main model equations and tables presented in the previous Section 2.

The optimization objective is the achievement of a 100 % renewable operation of the plant. This implies that, over the entire year, if no failures occur, the engine is not used to satisfy unmet energy related to the unavailability of energy or hydrogen in the storage systems. Thus, the storage systems should always be able to recover energy available in excess every time it is needed. For the storage systems, the following constraints have been set for the optimization on the initial state and on the minimum and maximum thresholds:

1) $SoC(0) = 95\%$, with $SoC_{min} = 20\%$ and $SoC_{max} = 95\%$
2) $TL(0) = 0m3$, with $TL_{min} = 0.1TL_{max}$ and TL_{max} 3m3.

The tank pressure has been set to 300 bar, and it is worth underlining that one unique tank has been considered to receive both the biogas produced by the digester and the hydrogen produced by the URSOFC in EL-mode. Therefore, in the final results, the tank size is given in cubic meters, and the mass of mixture available in the tank depends on the composition of the mixture itself, which is a function of both the control strategy and load profile.

In the full factorial design-based optimization, the PV-rated power has been spanned between 80 and 180 kW, while the URSOFC power in EL-mode has been varied between 80 kW and 150 kW. These thresholds have been set based on a pre-optimization screening that allowed to set the minimum power level for both components. The choice has been made to size the URSOFC based on the rated power in EL-mode since this is the highest rated power of such a system (i.e., for a given stack and a given current flow through the system, the maximum power in EL-mode is higher than the maximum power in FC-mode).

The resulting components sizes are provided in Table 4.

As one may note from Fig. 9, for Houston (a), New York (c), Las Vegas (e), Denver (f), and Los Angeles (g), the chosen plant sizing is the one that allows for a 100 % renewable operation, without excess energy. For Seattle, Fig. 8 (b), for Miami, Fig. 8 (d), and for Minneapolis, Fig. 8 (h), there is no solution in the considered optimization span that leads to no fuel usage and no excess energy. The smallest plant size that avoids fossil fuel use leads to a certain amount of excess energy to be dumped. At the same time, some solutions can nullify this excess energy, but without avoiding the usage of fossil fuel. Since the constraint on the 100 % renewable operation has been considered as a hard constraint and the constraint of zero excess energy, as a soft constraint,150-kW URSOFC and 150-kW PV have been chosen as the final sizing for Seattle and Miami, while 100-kW URSOFC and 120-kW PV have been chosen for Minneapolis.

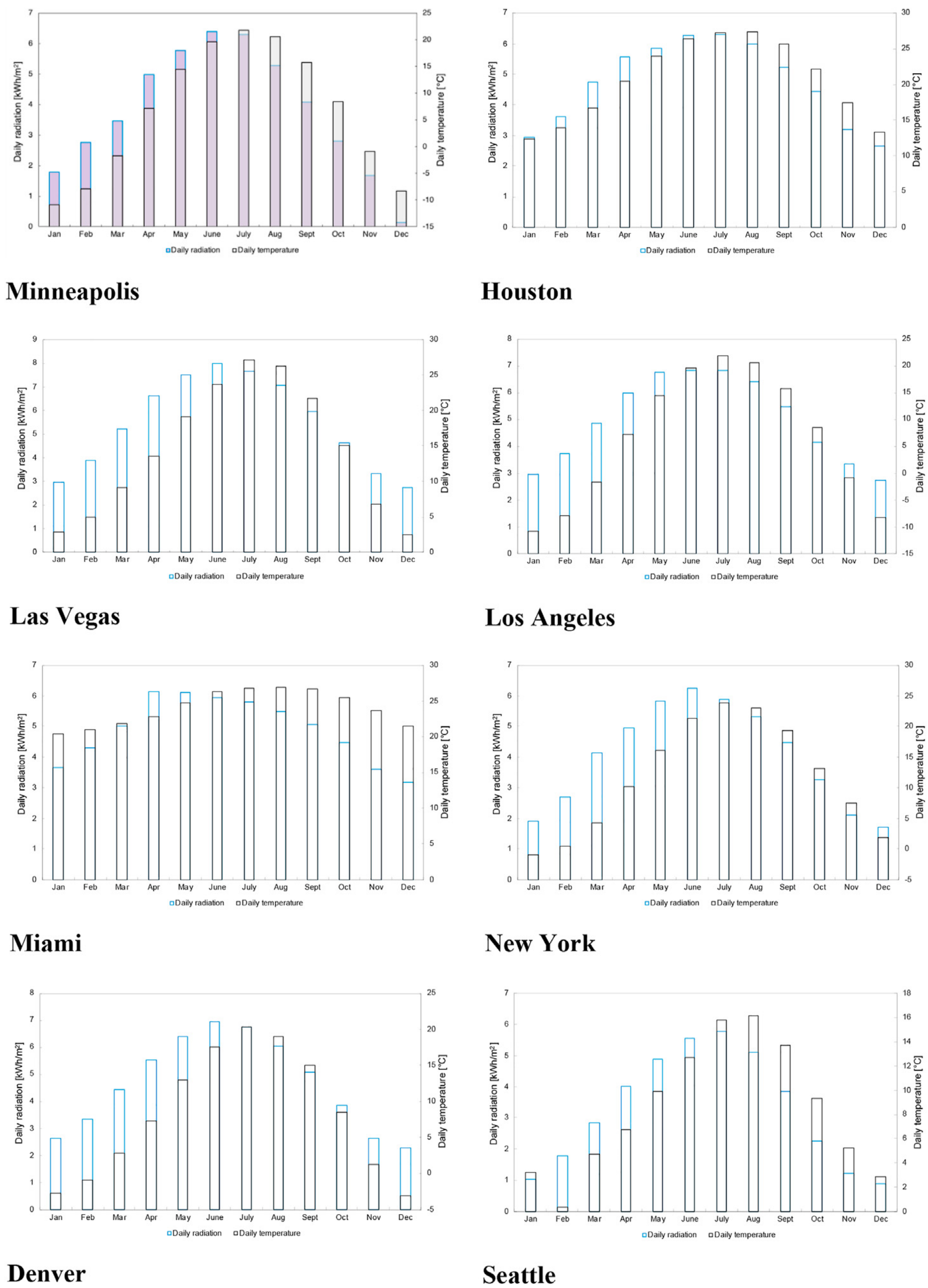


Fig. 5. Monthly average daily radiation and temperature values for each city.

3. Results and discussion

The results obtained from the simulations are presented and discussed for each of the eight locations. In order to consider the effect of different levels of stored energy, two different conditions have been here considered:

1) Halving of digester size.
2) $SoC(0) = 20\%$.

The simulation setup results in the four scenarios are summarized in Table 5.

The first scenario is used for the power plant sizing procedure and used as a reference.

The digester size reduction is useful to analyze its utilization factor. Halving the digester size allows having a digester utilization factor of around 100 % for all the locations, unlike the reference case, where the total biogas production all over the year never reaches the maximum digester capability (i.e., 10 800 std m^3 of produced biogas), for none of the locations.

As shown in Table 6, the reference configuration allows avoiding internal combustion engine usage while minimizing and often preventing excess energy waste through dump loads. Indeed, all the locations in Scenarios 2, 3, and 4 are characterized by almost zero excess energy, while this is not true for diesel fuel usage.

Fig. 9 portrays the total biogas production over one year for each location in Scenario 1, compared to Scenario 3. For both scenarios, Miami and New York (tropical and humid continental climates) are the locations with the highest biogas production, i.e., 9444 stdm^3 for Miami and 9093 stdm^3 for New York which means the highest

Table 3
Strip mall characteristics [15].

Building	
Name	Strip Mall
Activity	Commercial
Form	
Total Floor Area (m^2)	2090
Aspect Ratio	4
Number of Floors	1
Roof Type	Built-up flat roof, insulation entirely above deck
Exterior walls	
Construction Type	Steel-frame
Gross Dimensions - Total Area (m^2)	1184
Net Dimensions - Total Area (m^2)	1060
Wall to Skin Ratio	0.36
Roof	
Construction Type	IEAD
Gross Dimensions - Total Area (m^2)	2090
Net Dimensions - Total Area (m^2)	2090
Roof to Skin Ratio	0.64

digester utilization factor. Instead, the effect of the different initial state of charge is relevant only for Seattle and Las Vegas, as also confirmed by Fig. 10 (a) and (b), that show the daily biogas production averaged on a monthly basis for the eight locations in Scenarios 1 and 3.

Recalling that the biogas production rate by the digester is subject to the value of the tank level, it is worth pointing out that the particular behavior in January, see Fig. 10 (a) and (b) and also confirmed in Scenarios 2 and 4 as well, is mainly driven by the initial condition of an empty tank. Therefore, the digester utilization factor is always 100 % during this month, except for two cities, Las Vegas and Seattle. The different behavior in the two scenarios for these two cities can be better caught by looking at the biogas tank level, and battery SoC level provided for Scenario 1 and 3, only for Las Vegas, in Fig. 11.

As one can note from Fig. 11, in Scenario 1, the control strategy detects an initial SoC (i.e., 95 %) higher than SoC_{SUST} (i.e., 50 %) and selects the first “Power Recovery” mode. Therefore, the URSOFC is used as the primary backup system (EL-mode), and, on average, the tank is replenished faster than in Scenario 3 throughout the year. On the other hand, this implies a lower biogas concentration in the tank, on average, which in turn makes the URSOFC work in FC-mode with a higher available power, also emptying the tank at a faster rate (see, for example, in May, in Fig. 11). The first effect can be seen in lower biogas production for Scenario 1, in February, April, and November, compare Fig. 10 (a) and (b), since the tank is replenished faster, while the second effect can be seen in June, October or December when more biogas is produced in Scenario 1 to replenish the tank that is being emptied faster.

Similar conclusions may be drawn for Seattle, but they are not reported here for the sake of fluency.

Figs. 10 and 11 show the utilization factor of 100 % for the digester from June to August and then again from November to March for New York and from June to November, and again in January for Miami.

Fig. 12 shows that New York has a PV output/load ratio comparable to Miami. Still, Miami, having a tropical climate, has a higher amount of extra energy to be stored during winter. In contrast, New York, having a humid continental climate, has a higher amount of extra energy to be stored during summer. This results in the behavior of Fig. 13 (a) and (b) and Fig. 14 (a) and (b). For New York, the power in EL-mode is lower than in Miami during winter due to the lower amount of energy to be recovered.

In comparison, the power in FC-mode is lower than in Miami during summer due to the higher amount of energy available from the PV panels. In Fig. 12, it is also interesting to note that, for New York city, at around time 5000 h, the ratio PV Output/Load Energy is much lower than one (PV Output lower than load). Fig. 13 (a) and (b) show that, for New York, being the maximum power of the URSOFC lower than in Miami, this asks for the battery usage to meet the load power demand. Therefore, the battery is discharged below SOC_{SUST} equal to 50 %, and the Power Recovery b) mode is activated until the battery charge is completely restored. Fig. 12 shows that, also for Miami, at around time 5000 h, the ratio PV Output/Load Energy is much lower than one.

Nonetheless, the URSOFC has higher rated power. Thus, the URSOFC alone can meet the load power demand. However, the tank level is lower than in New York, mainly because of the higher mass flow rate consumption of the URSOFC in FC-mode and the lower extra power to be absorbed in EL-mode.

At the same time, Seattle (Temperate climate) has the lowest biogas production, thus the lowest digester utilization factor. This is due to the lower electric demand compared to the available daily radiation, which results in a high PV output/load ratio, which implies a large use of the backup systems (see Fig. 15 (a) and (b)). This way the tank is often replenished by the URSOFC in EL-mode, and the digester must be switched off. This, in turn, results also in a mixture contained in the storage tank, with lower biogas concentration and thus higher FC efficiencies.

Fig. 16 shows that the digester utilization factor increases if the amount of biogas produced daily is halved since the biogas production is insufficient to replenish the tank and the digester is never (or very seldom, i.e., only for Seattle) switched off. Nonetheless, the total amount of biogas produced at the end of the year is significantly reduced, reaching a maximum value of 5475 $stdm^3$ (see Fig. 16). This reflects an identical daily average of biogas produced by the digester monthly for each location.

Since the tank is empty in all the scenarios, for all the locations, at the beginning of the year, the main effect of the reduction of the state of charge is the unavailability of energy during the first hours of the night of the first day of the year. This, leads to an unavoidable usage of the internal combustion engine to meet the load power request, until the PV output becomes higher than the load, which happens after 32 h (8:00 a.m. of January 2) to 36 h (10:00 a.m. of January 2) depending on the location. Nevertheless, after the first hour, some biogas is already available, and the URSOFC can be used to reduce the diesel power request. In fact, Fig. 17 shows that the diesel engine runs for no more than 26 h in every location. From Fig. 17, one can see that this leads to fuel consumption that goes from a minimum of 97.9 kg in Houston to a maximum of 180 kg in

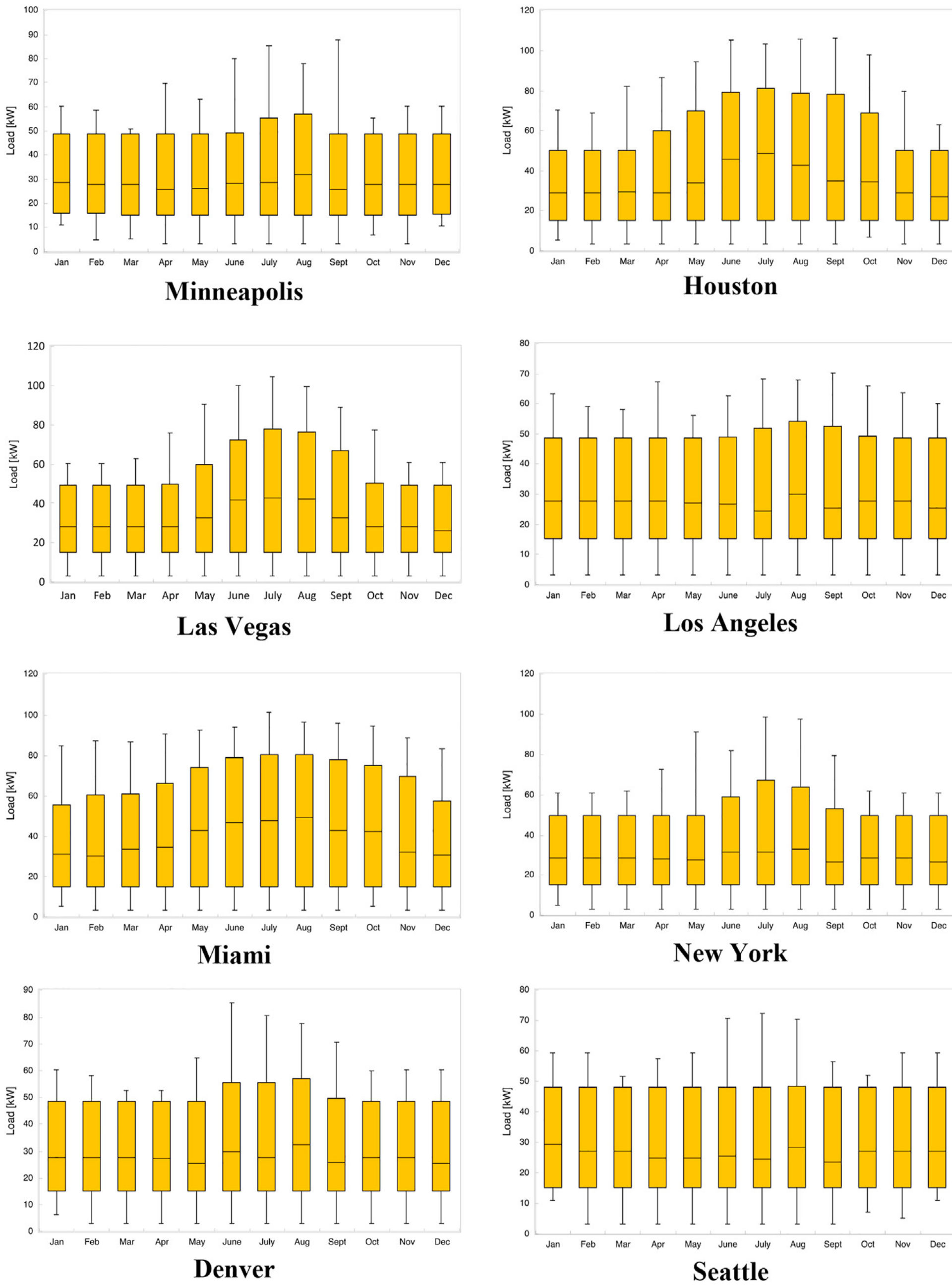


Fig. 6. Strip mall monthly electrical demand in different cities.

Miami if the biogas production is not halved. Obviously, the worst-case scenario is the fourth one, where the biogas production is reduced, and the battery is initially not charged. Consequently, the fuel consumption goes from a minimum of 6351 kg in Seattle to a maximum of 16.7 tons in Miami, with a significant increase in the diesel engine operating time (Fig. 18).

For the sake of fluency, all the data of the most representative figures for this section not already present in the text have been placed in a dedicated Appendix.

4. Conclusions

In this paper, an off-grid hybrid RES power plant with various energy storage systems was designed for commercial building application, and its operation was investigated considering the sizing and energy management under different U.S. climate zones.

In the proposed power plant, a PV system was assumed to be a primary energy supplier, while the URSOFC, together with a battery and a diesel generator, were employed as a backup system to cope with the intermittency of the main power source.

The use of biogas, composed of CH_4/CO_2 to feed a reversible solid oxide fuel cell has also been assessed, reproducing a detailed electrochemical model of the URSOFC both in electrolyzer and fuel cell modes. Such a model estimates the system performance taking into account the balance of power of all the needed auxiliary systems.

Based on different amounts of biogas produced by a food-waste digester, and different state of charge of batteries, four different scenarios have been simulated and analyzed for the eight locations. The optimization target - 100 % renewable operation without

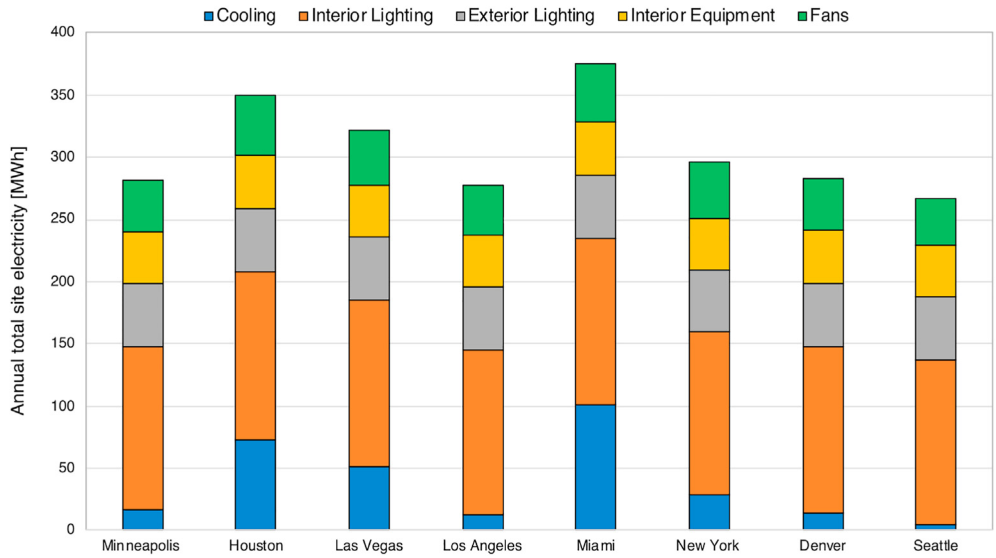


Fig. 7. Annual total site electricity for each end-uses.

Table 4
Components sizes.

City	URSOFC [kW]	PV [kW]	ICE [kW]	BAT [–]	DIG [m^3]	TANK [m^3]
Houston	150	150	120	5S 1P	87	3
Seattle	150	150	80	5S 1P	87	3
New York	120	120	110	5S 1P	87	3
Miami	150	150	110	5S 1P	87	3
Las Vegas	120	150	110	5S 1P	87	3
Denver	100	100	90	5S 1P	87	3
Los Angeles	100	100	80	5S 1P	87	3
Minneapolis	100	120	100	5S 1P	87	3

excess energy - is achieved only if the digester produces from 6000 stdm^3/y up to 9500 stdm^3/y, depending on the climatic zone.

Based on the study's main outcomes, the reference configuration allows avoiding the internal combustion engine usage while minimizing and often preventing the waste of excess energy through dump loads. However, it should be mentioned that by reducing the biogas availability or starting with a low state of charge, the use of the diesel generator backup system cannot be avoided.

Future studies will consider the implementation, in the control strategy, of a constraint on the state of charge at the end of the year, minimizing or nullifying the diesel fuel usage. Moreover, an economic and environmental assessment of the proposed system will be conducted to estimate the cost of energy and the initial investment for this kind of solution.

In conclusion, the methodology presented here provides a template that may be applied to other sites and buildings interested in quantifying the energy, economic, and resiliency benefits of RES.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Barbara Mendecka: Investigation, Data curation, Validation, Writing – original draft, Writing – review & editing. **Daniele Chiappini:** Conceptualization, Methodology, Formal analysis, Resources, Investigation, Data curation, Validation, Writing – original draft, Writing – review & editing. **Laura Tribioli:** Conceptualization, Methodology, Formal analysis, Resources, Investigation, Data curation, Validation, Writing – original draft, Writing – review & editing. **Raffaello Cozzolino:** Conceptualization, Methodology, Formal analysis, Resources, Investigation, Data curation, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

In this section, the results presented in Section 3 will be enumerated in summarizing tables related to the most important Figures.

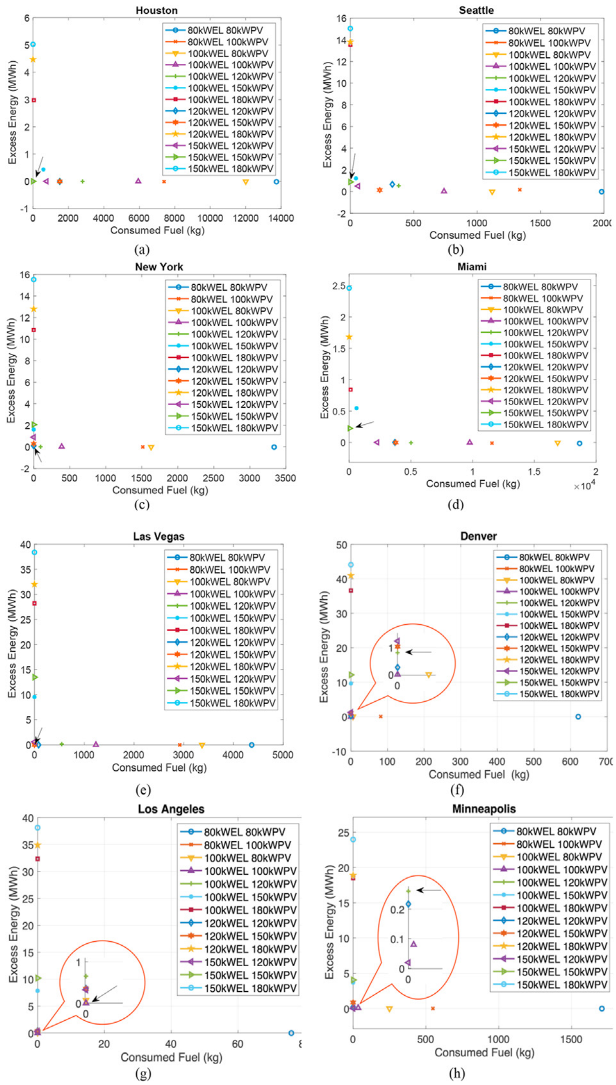


Fig. 8. Distribution of the different URSOFC and PV sizes in the Excess Energy-Consumed Fuel space for the full factorial optimization for Houston (a), Seattle (b), New York (c), Miami (d), Las Vegas (e), Denver (f), Los Angeles (g), Minneapolis (h).

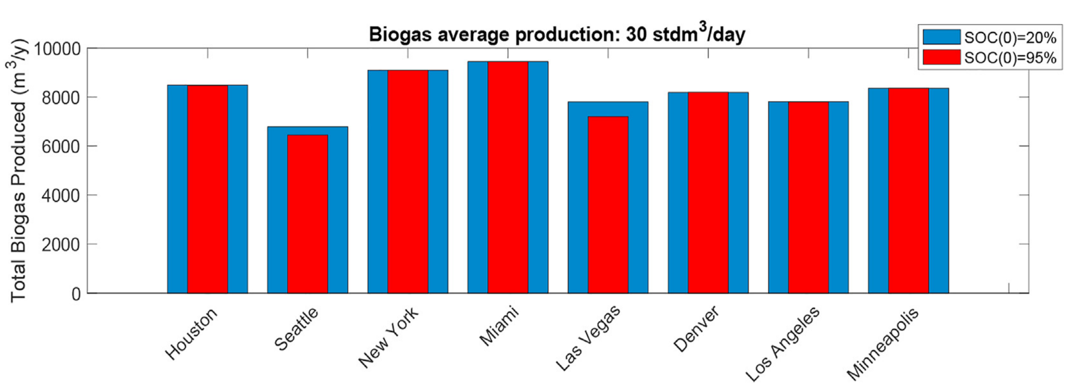


Fig. 9. Total biogas produced by the digester on yearly basis for each location – Scenario 1 (red) vs Scenario 3 (blue).

Table 5
Summary of the simulated scenarios.

No. of the Scenario	Biogas Average Production [std m^3/day]	Initial State of Charge [%]
1	30	95
2	15	95
3	30	20
4	15	20

Table 6
Fossil fuel yearly consumption in the two scenarios for each location.

No. of the Scenario	Diesel Average Consumption [kg]							
	Houston	Seattle	New York	Miami	Las Vegas	Denver	Los Angeles	Minneapolis
1	0	0	0	0	0	0	0	0
2	10863	5757	12732	14811	6792	101171	6813	10467
3	97.9	174	172	180	125	131	119	172
4	11056	6351	12916	14977	6984	10356	6991	10649

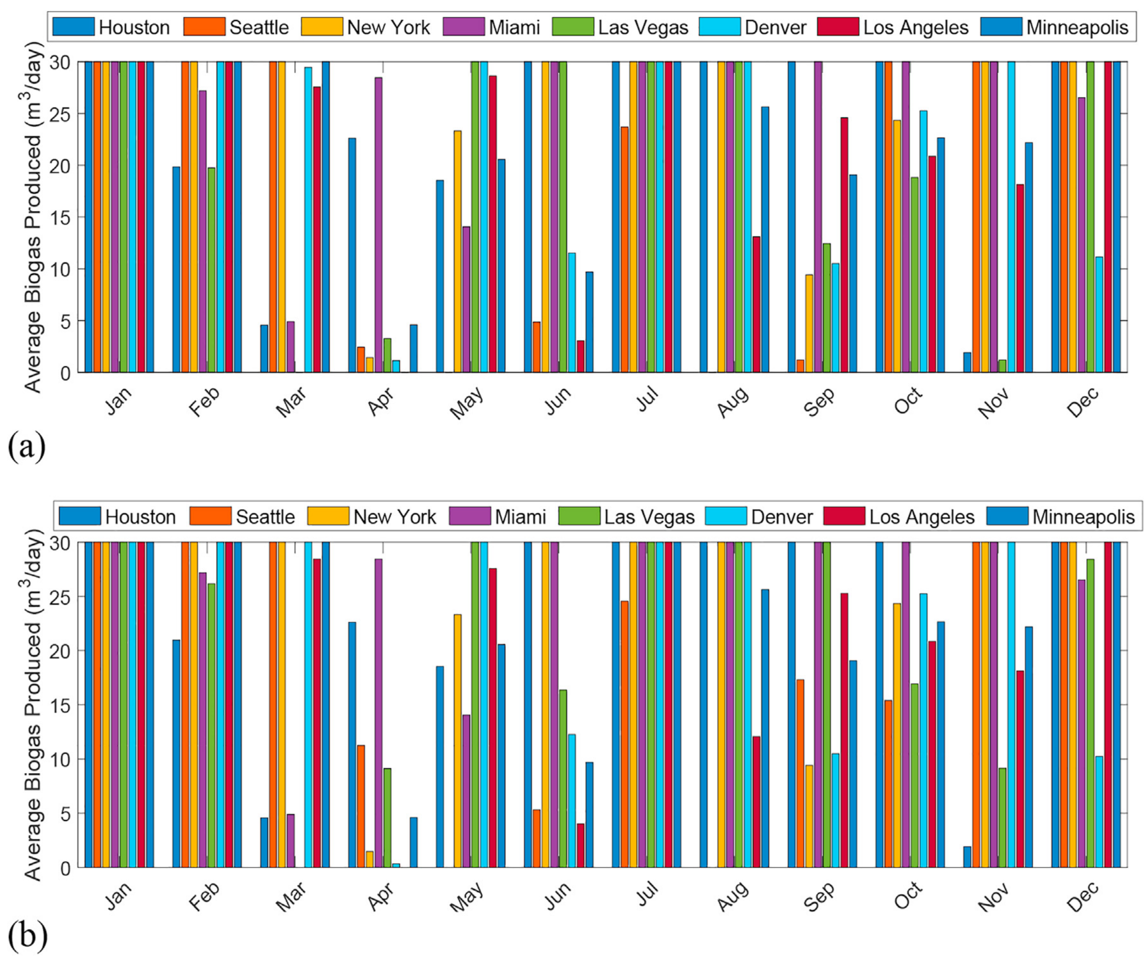


Fig. 10. Daily average of biogas produced by the digester on monthly basis for each location – Scenario 1 (a) vs Scenario 3 (b).

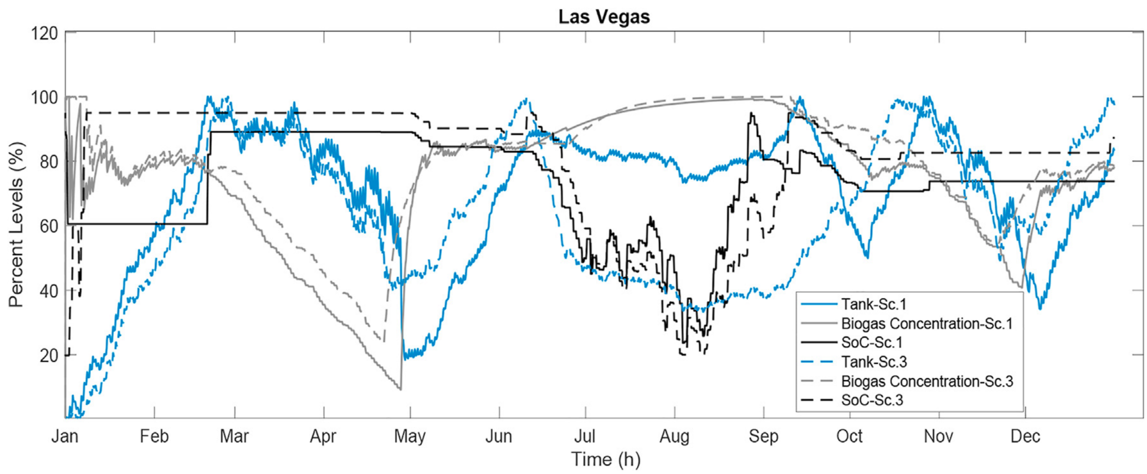


Fig. 11. Tank and SoC levels for Las Vegas – Scenario 1 (solid line) vs Scenario 3 (dashed line).

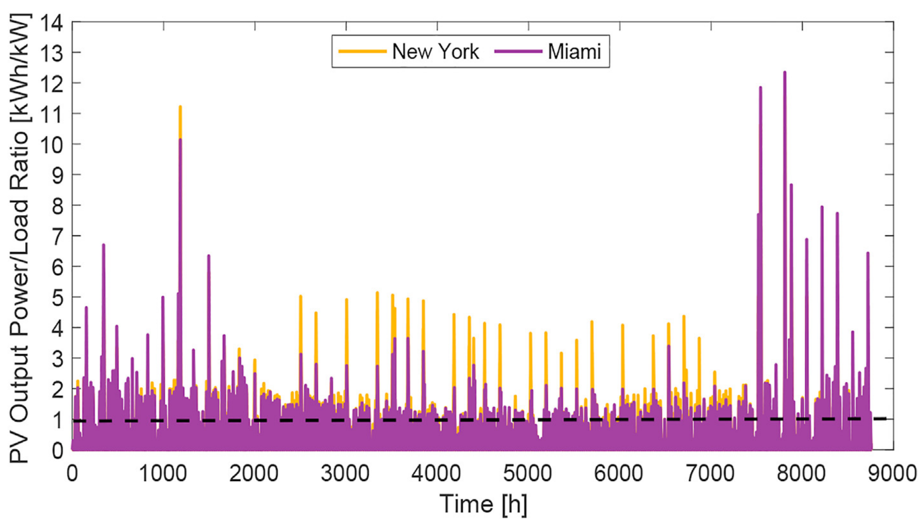


Fig. 12. PV Output Energy to Load Energy ratio for New York and Miami – Scenario 1.

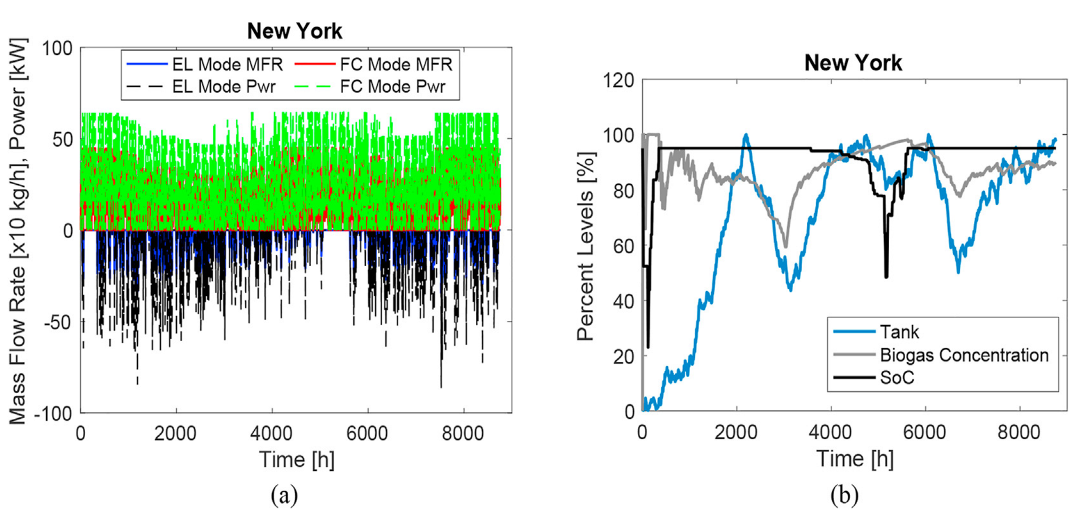


Fig. 13. (a) Mass flow rate (MFR) and Power (Pwr) of the URSOFC for New York - Scenario 1; (b) Tank and SOC levels and biogas concentration for New York - Scenario 1.

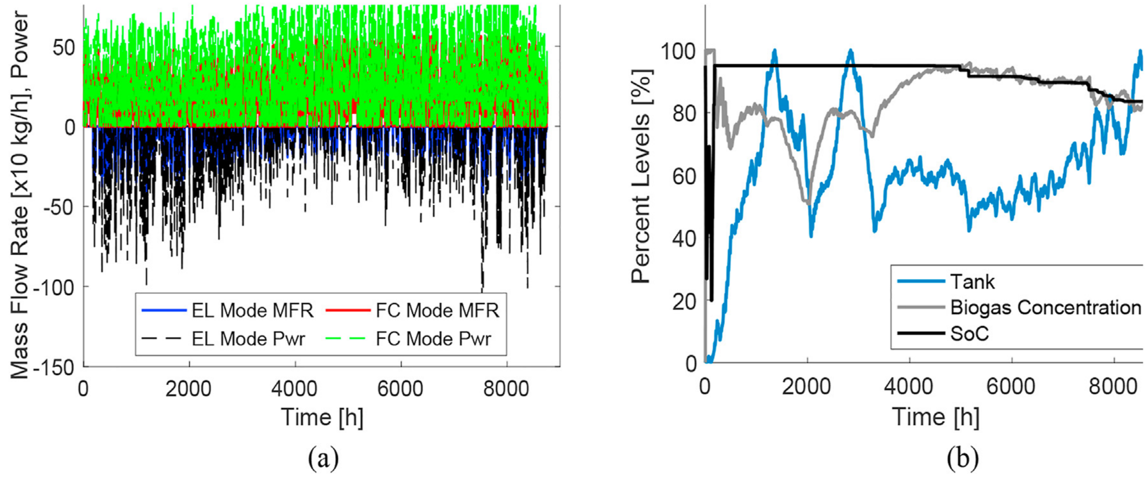


Fig. 14. (a) Mass flow rate (MFR) and Power (Pwr) of the URSOFC for Miami - Scenario 1; (b) Tank and SOC levels and biogas concentration for Miami - Scenario 1.

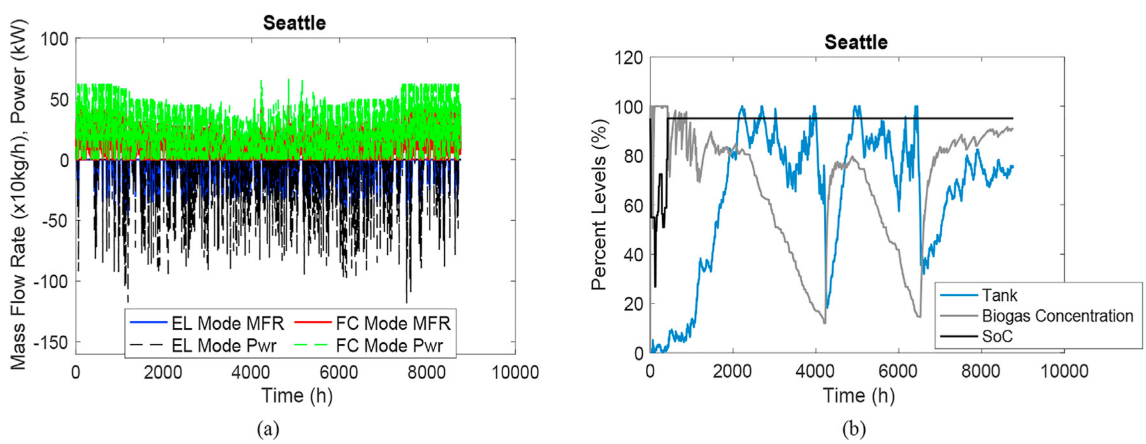


Fig. 15. (a) Mass flow rate (MFR) and Power (Pwr) of the URSOFC for Seattle - Scenario 1; (b) Tank and SOC levels and biogas concentration for Seattle - Scenario 1.

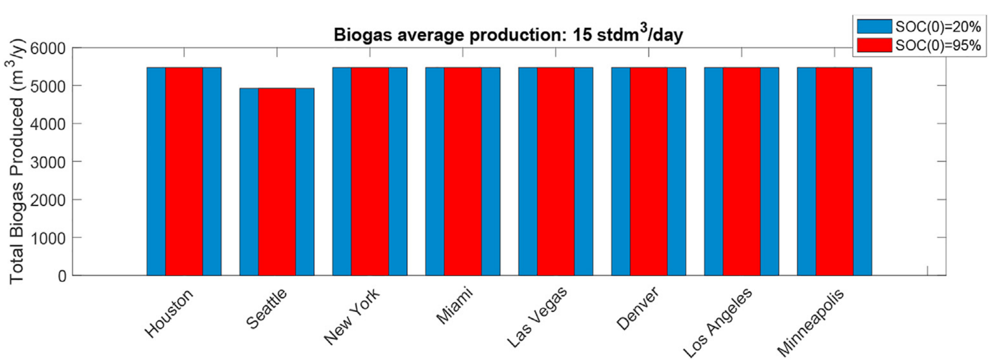


Fig. 16. Total biogas produced by the digester on a yearly basis for each location – Scenario 2 (red) vs. Scenario 4 (blue).

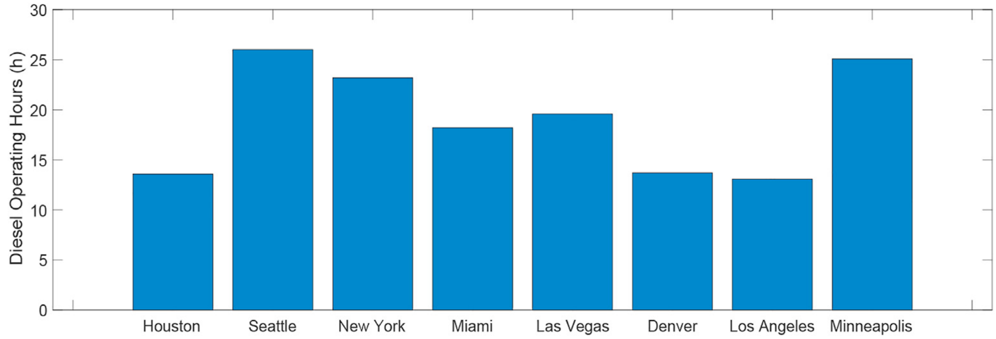


Fig. 17. Diesel Engine operating hours over the year in the eight locations – Scenario 3.

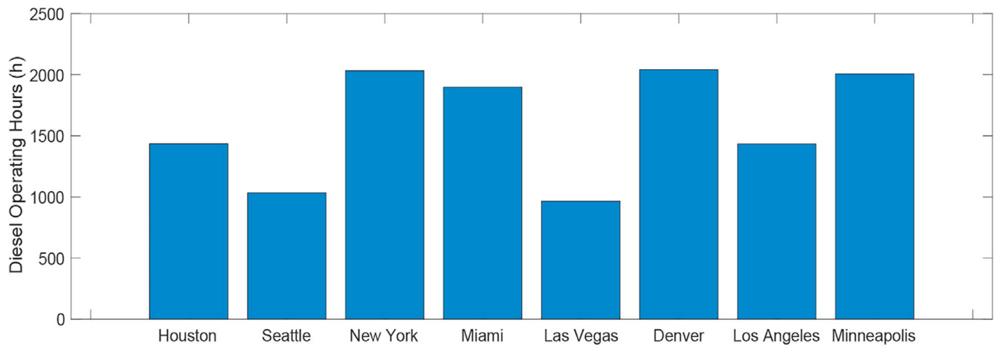
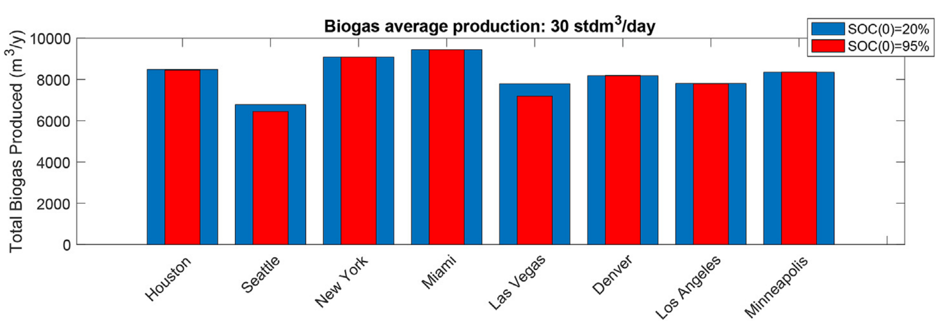


Fig. 18. Diesel Engine operating hours over the year in the eight locations – Scenario 4.

Table A. 1
Data from Fig. 9



	Scenario 1	Scenario 3
Houston	8457	8490
Seattle	6449	6785
New York	9093	9094
Miami	9444	9444
Las Vegas	7193	7798
Denver	8197	8182
Los Angeles	7796	7810
Minneapolis	8360	8360

Table A. 2
Data from Fig. 10 (a)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
Houston	30.00	19.84	4.57	22.61	18.54	30.00	30.00	30.00	30.00	30.00	1.91
Seattle	30.00	30.00	30.00	2.43	0.00	4.87	23.70	0.00	1.19	30.00	30.00
New York	30.00	30.00	30.00	1.43	23.31	30.00	30.00	30.00	9.41	24.34	30.00
Miami	30.00	27.18	4.90	28.45	14.04	30.00	30.00	30.00	30.00	30.00	30.00
Las Vegas	30.00	19.74	0.00	3.27	30.00	30.00	30.00	30.00	12.43	18.81	1.19
Denver	30.00	30.00	29.46	1.15	30.00	11.52	30.00	30.00	10.51	25.25	30.00
Los Angeles	30.00	30.00	27.55	0.00	28.63	3.04	30.00	13.08	24.57	20.86	18.13
Minneapolis	30.00	30.00	30.00	4.59	20.56	9.69	30.00	28.63	19.07	22.64	22.18

Table A. 3
Data from Fig. 10 (b)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
Houston	30.00	20.97	4.58	22.62	18.54	30.00	30.00	30.00	30.00	30.00	1.91
Seattle	30.00	30.00	30.00	11.26	0.00	5.31	24.56	0.00	17.30	15.39	30.00
New York	30.00	30.00	30.00	1.47	23.32	30.00	30.00	30.00	9.41	24.34	30.00
Miami	30.00	27.18	4.90	28.45	14.04	30.00	30.00	30.00	30.00	30.00	30.00
Las Vegas	30.00	26.17	0.00	9.13	30.00	16.36	30.00	30.00	30.00	16.93	9.16
Denver	30.00	30.00	30.00	0.32	30.00	12.27	30.00	30.00	10.49	25.24	30.00
Los Angeles	30.00	30.00	28.45	0.00	27.58	4.01	30.00	12.07	25.28	20.85	18.12
Minneapolis	30.00	30.00	30.00	4.60	20.56	9.69	30.00	25.64	19.07	22.64	22.18

Table A. 4
Data from Fig. 16

	Scenario 2	Scenario 4
Houston	5475	5475
Seattle	4927	4927
New York	5475	5475
Miami	5475	5475
Las Vegas	5475	5475
Denver	5475	5475
Los Angeles	5475	5475
Minneapolis	5475	5475

References

[1] IEA and UNEP, 2019 Global Status Report for Buildings and Construction, 2019.
[2] U.S. Energy Information Administration, January 2021 Monthly Energy Review, 2021.
[3] Energy U.S. Department, Commercial buildings energy consumption survey, U.S. Energy Inf. Adm. (2018) 1–269.
[4] J. Kneifel, Life-cycle Carbon and Cost Analysis of Energy Efficiency Measures in New Commercial Buildings, Energy Build, 2010, https://doi.org/10.1016/j.enbuild.2009.09.011.
[5] J. Kneifel, Beyond the Code: Energy, Carbon, and Cost Savings Using Conventional Technologies, Energy Build, 2011, https://doi.org/10.1016/j.enbuild.2010.12.019.
[6] E. Barnes, K. Parrish, Small buildings, big impacts: developing a library of small commercial building energy efficiency case studies, in: Procedia Eng., Elsevier Ltd, 2015, pp. 853–860, https://doi.org/10.1016/j.proeng.2015.08.523.
[7] N. Fernandez, S. Katipamula, W. Wang, Y. Xie, M. Zhao, Energy savings potential from improved building controls for the US commercial building sector, Energy Effic. 11 (2018) 393–413, https://doi.org/10.1007/s12053-017-9569-5.
[8] L. Mariam, M. Basu, M.F. Conlon, Microgrid: architecture, policy and future trends, Renew. Sustain. Energy Rev. (2016), https://doi.org/10.1016/j.rser.2016.06.037.
[9] M.H. Mostafa, S.H.E. Abdel Aleem, S.G. Ali, A.Y. Abdelaziz, Energy-management solutions for microgrids, in: Distrib. Energy Resour. Microgrids Integr. Challenges Optim., Elsevier, 2019, pp. 483–515, https://doi.org/10.1016/B978-0-12-817774-7.00020-X.
[10] A.K. Pandey, V.V. Tyagi, J.A. Selvaraj, N.A. Rahim, S.K. Tyagi, Recent advances in solar photovoltaic systems for emerging trends and advanced applications, Renew. Sustain. Energy Rev. (2016), https://doi.org/10.1016/j.rser.2015.09.043.
[11] B. Griffith, P. Torcellini, N. Long, D. Crawley, J. Ryan, Assessment of the Technical Potential for Achieving Zero-Energy Commercial Buildings, 2006.
[12] B. Mendecka, R. Cozzolino, M. Leveni, G. Bella, Energetic and exergetic performance evaluation of a solar cooling and heating system assisted with thermal storage, Energy 176 (2019) 816–829, https://doi.org/10.1016/j.energy.2019.04.024.
[13] M.D.A. Al-falahi, S.D.G. Jayasinghe, H. Enshaei, A review on recent size optimization methodologies for standalone solar and wind hybrid renewable energy system, Energy Convers. Manag. 143 (2017) 252–274, https://doi.org/10.1016/j.enconman.2017.04.019.
[14] L. Tribioli, R. Cozzolino, L. Evangelisti, G. Bella, Energy management of an off-grid hybrid power plant with multiple energy storage systems, Energies 9 (2016) 661, https://doi.org/10.3390/en9080661.
[15] L. Tribioli, R. Cozzolino, Techno-economic analysis of a standalone microgrid for a commercial building in eight different climate zones, Energy Convers. Manag. 179 (2019) 58–71, https://doi.org/10.1016/j.enconman.2018.10.061.
[16] E.I. Come Zebra, H.J. van der Windt, G. Nhumaio, A.P.C. Faaij, A review of hybrid renewable energy systems in mini-grids for off-grid electrification in developing countries, Renew. Sustain. Energy Rev. 144 (2021), 111036, https://doi.org/10.1016/j.rser.2021.111036.
[17] J.M. Aberilla, A. Gallego-Schmid, L. Stamford, A. Azapagic, Design and environmental sustainability assessment of small-scale off-grid energy systems for remote rural communities, Appl. Energy (2020), https://doi.org/10.1016/j.apenergy.2019.114004.
[18] U. Datta, A. Kalam, J. Shi, Hybrid PV-wind renewable energy sources for microgrid application: an overview, in: Hybrid-Renewable Energy Syst. Microgrids Integr. Dev. Control, 2018, https://doi.org/10.1016/B978-0-08-102493-5.00001-7.
[19] S. Buragohain, K. Mohanty, P. Mahanta, Hybridization of solar photovoltaic and biogas system: experimental, economic and environmental analysis, Sustain. Energy Technol. Assessments. 45 (2021), 101050, https://doi.org/10.1016/j.seta.2021.101050.
[20] J.J.D. Nesamalar, S. Suruthi, S.C. Raja, K. Tamilarasu, Techno-economic analysis of both on-grid and off-grid hybrid energy system with sensitivity analysis for an educational institution, Energy Convers. Manag. 239 (2021), 114188, https://doi.org/10.1016/j.enconman.2021.114188.
[21] L. Ji, X. Liang, Y. Xie, G. Huang, B. Wang, Optimal design and sensitivity analysis of the standalone hybrid energy system with PV and biomass-CHP for remote villages, Energy 225 (2021), 120323, https://doi.org/10.1016/j.energy.2021.120323.
[22] F. Kahwash, A. Maheri, K. Mahkamov, Integration and optimisation of high-penetration Hybrid Renewable Energy Systems for fulfilling electrical and thermal demand for off-grid communities, Energy Convers. Manag. 236 (2021), https://doi.org/10.1016/j.enconman.2021.114035.
[23] A. Allouhi, S. Rehman, M. Krarti, Role of energy efficiency measures and hybrid PV/Biomass power generation in designing 100% electric rural houses: a case study in Morocco, Energy Build. (2021), 110770, https://doi.org/10.1016/j.enbuild.2021.110770.
[24] L. Al-Ghussain, A. Darwish Ahmad, A.M. Abubaker, M.A. Mohamed, An integrated photovoltaic/wind/biomass and hybrid energy storage systems towards 100% renewable energy microgrids in university campuses, Sustain. Energy Technol. Assessments. 46 (2021), 101273, https://doi.org/10.1016/j.seta.2021.101273.
[25] A. Cano, P. Arévalo, F. Jurado, Energy analysis and techno-economic assessment of a hybrid PV/HKT/BAT system using biomass gasifier: Cuenca-Ecuador case study, Energy 202 (2020), 117727, https://doi.org/10.1016/

j.energy.2020.117727.
[26] S. Vendoti, M. Muralidhar, R. Kiranmayi, Techno-economic analysis of off-grid solar/wind/biogas/biomass/fuel cell/battery system for electrification in a cluster of villages by HOMER software, Environ, Dev. Sustain. 23 (2021) 351–372, https://doi.org/10.1007/s10668-019-00583-2.
[27] F. Dawood, G.M. Shafiullah, M. Anda, Stand-alone microgrid with 100% renewable energy: a case study with hybrid solar pv-battery-hydrogen, Sustain. Times 12 (2020) 2047, https://doi.org/10.3390/su12052047.
[28] M.R. Elkadeem, S. Wang, S.W. Sharshir, E.G. Atia, Feasibility analysis and techno-economic design of grid-isolated hybrid renewable energy system for electrification of agriculture and irrigation area: a case study in Dongola, Sudan, Energy Convers. Manag. 196 (2019) 1453–1478, https://doi.org/10.1016/j.enconman.2019.06.085.
[29] T. Sarkar, A. Bhattacharjee, H. Samanta, K. Bhattacharya, H. Saha, Optimal design and implementation of solar PV-wind-biogas-VRFB storage integrated smart hybrid microgrid for ensuring zero loss of power supply probability, Energy Convers. Manag. (2019), https://doi.org/10.1016/j.enconman.2019.04.025.
[30] L.M. Halabi, S. Mekhilef, Flexible hybrid renewable energy system design for a typical remote village located in tropical climate, J. Clean. Prod. 177 (2018) 908–924, https://doi.org/10.1016/j.jclepro.2017.12.248.
[31] A.C. Duman, Ö. Güler, Techno-economic analysis of off-grid PV/wind/fuel cell hybrid system combinations with a comparison of regularly and seasonally occupied households, Sustain, Cities Soc. 42 (2018) 107–126, https://doi.org/10.1016/j.scs.2018.06.029.
[32] B.K. Das, N. Hoque, S. Mandal, T.K. Pal, M.A. Raihan, A techno-economic feasibility of a standalone hybrid power generation for remote area application in Bangladesh, Energy 134 (2017) 775–788, https://doi.org/10.1016/j.energy.2017.06.024.
[33] R. Rajbongshi, D. Borgohain, S. Mahapatra, Optimization of PV-biomass-diesel and grid base hybrid energy systems for rural electrification by using HOMER, Energy 126 (2017) 461–474, https://doi.org/10.1016/j.energy.2017.03.056.
[34] M. Hossain, S. Mekhilef, L. Olatomiwa, Performance evaluation of a standalone PV-wind-diesel-battery hybrid system feasible for a large resort center in South China Sea, Malaysia, Sustain. Cities Soc. 28 (2017) 358–366, https://doi.org/10.1016/j.scs.2016.10.008.
[35] J. Ahmad, M. Imran, A. Khalid, W. Iqbal, S.R. Ashraf, M. Adnan, S.F. Ali, K.S. Khokhar, Techno economic analysis of a wind-photovoltaic-biomass hybrid renewable energy system for rural electrification: a case study of Kallar Kahar, Energy 148 (2018) 208–234, https://doi.org/10.1016/j.energy.2018.01.133.
[36] L. Tribioli, R. Cozzolino, L. Evangelisti, G. Bella, L. Tribioli, R. Cozzolino, L. Evangelisti, G. Bella, Energy management of an off-grid hybrid power plant with multiple energy storage systems, Energies 9 (2016) 661, https://doi.org/10.3390/en9080661.
[37] R. Cozzolino, L. Tribioli, G. Bella, Power management of a hybrid renewable system for artificial islands: a case study, Energy 106 (2016) 774–789, https://doi.org/10.1016/j.energy.2015.12.118.
[38] A. Heydari, A. Askarzadeh, Techno-economic analysis of a PV/biomass/fuel cell energy system considering different fuel cell system initial capital costs, Sol. Energy (2016), https://doi.org/10.1016/j.solener.2016.04.018.
[39] R. Cozzolino, L. Lombardi, L. Tribioli, Use of biogas from biowaste in a solid oxide fuel cell stack: application to an off-grid power plant, Renew. Energy 111 (2017) 781–791, https://doi.org/10.1016/j.renene.2017.04.027.
[40] Y. Sawle, S.C. Gupta, A.K. Bohre, Review of hybrid renewable energy systems with comparative analysis of off-grid hybrid system, Renew. Sustain. Energy Rev. 81 (2018) 2217–2235, https://doi.org/10.1016/j.rser.2017.06.033.
[41] Z. Abdin, A. Zafaranloo, A. Rafiee, W. Mérida, W. Lipiński, K.R. Khalilpour, Hydrogen as an energy vector, Renew. Sustain. Energy Rev. (2020), https://doi.org/10.1016/j.rser.2019.109620.
[42] G. Tzamalis, E.I. Zoulias, E. Stamatakis, E. Varkaraki, E. Lois, F. Zannikos, Techno-economic analysis of an autonomous power system integrating hydrogen technology as energy storage medium, Renew. Energy (2011), https://doi.org/10.1016/j.renene.2010.06.006.
[43] S.N. Laboratory, Long-duration energy storage, ISSUE Br (2021).
[44] H. Ibrahim, A. Ilinca, J. Perron, Energy storage systems-Characteristics and comparisons, Renew. Sustain. Energy Rev. 12 (2008) 1221–1250, https://doi.org/10.1016/j.rser.2007.01.023.
[45] A.L. Facci, V. Cigolotti, E. Jannelli, S. Ubertini, Technical and economic assessment of a SOFC-based energy system for combined cooling, heating and power, Appl. Energy (2017), https://doi.org/10.1016/j.apenergy.2016.06.105.
[46] B. Mendecka, L. Tribioli, R. Cozzolino, Life Cycle Assessment of a standalone solar-based polygeneration power plant for a commercial building in different climate zones, Renew. Energy 154 (2020) 1132–1143, https://doi.org/10.1016/j.renene.2020.03.063.
[47] R. Cozzolino, D. Chiappini, L. Tribioli, Off-grid PV/URFC power plant fueled with biogas from food waste: an energetic and economic analysis, Energy 219 (2021), 119537, https://doi.org/10.1016/j.energy.2020.119537.
[48] A.A. Trendewicz, R.J. Braun, Techno-economic analysis of solid oxide fuel cell-based combined heat and power systems for biogas utilization at wastewater treatment facilities, J. Power Sources 233 (2013) 380–393, https://doi.org/10.1016/j.jpowsour.2013.01.017.
[49] D. Chiappini, A.L. Facci, L. Tribioli, S. Ubertini, SOFC management in distributed energy systems, J. Fuel Cell Sci. Technol. 8 (2011), https://doi.org/10.1115/1.4002907.
[50] R. Cozzolino, L. Tribioli, G. Bella, Power management of a hybrid renewable system for artificial islands: a case study, Energy 106 (2016) 774–789, https://doi.org/10.1016/j.energy.2015.12.118.
[51] Commercial Reference Buildings | Office of Energy Efficiency & Renewable Energy. Department of Energy (DOE). Available from:, (n.d.). https://www.energy.gov/eere/buildings/commercial-reference-buildings (accessed February 13, 2019).
[52] M.C. Baechler, T.L. Gilbride, P.C. Cole, M.G. Hefty, K. Ruiz, Guide to determining climate Regions by County, Pacific Northwest Natl, Lab. Oak Ridge Natl. Lab. 7 (2015) 1–34. https://www.energy.gov/sites/prod/files/2015/10/f27/ba_climate_region_guide_7.3.pdf.
[53] NASA Surface meteorology and Solar Energy database., Global horizontal radiation, monthly averaged values over 22-year period (July 1983–June 2005). Available from: ≤https://eosweb.larc.nasa.gov>, (n.d.).
[54] Homer Energy, Homer - hybrid renewable and distributed generation system design software, Http://Www.Homerenergy.Com, http://homerenergy.com, 2016.
[55] Y. Yang, W. Bao, G.H. Xie, Estimate of restaurant food waste and its biogas production potential in China, J. Clean. Prod. 211 (2019) 309–320, https://doi.org/10.1016/j.jclepro.2018.11.160.